Project 2
Part 1

1a Strategy

- When normalised, it should be the case that $\int_{-\infty}^{\infty} |\Psi(x,t)|^2 dx = 1$. Due to the conservation of probability, $\int_{-\infty}^{\infty} |\Psi(x,0)|^2 dx = 1$ implies that $\int_{-\infty}^{\infty} |\Psi(x,t)|^2 dx = 1$.
- Thus, we will focus on finding restrictions on A and a given that $\int_{-\infty}^{\infty} |\Psi(x,0)|^2 dx = 1$

Solution

$$\int_{-\infty}^{\infty} |\Psi(x,0)|^2 dx = \int_{-\infty}^{\infty} Ae^{-ax^2} \cdot Ae^{-ax^2} dx = A^2 \int_{-\infty}^{\infty} e^{-2ax^2} dx = 1$$

- Now, let $u = \sqrt{2a}\, x$, so $du = \sqrt{2a}\, dx \Rightarrow dx = \frac{1}{\sqrt{2a}} du$

$$\Rightarrow A^2 \int_{x=-\infty}^{x=\infty} \frac{1}{\sqrt{2a}} e^{-u^2} du = \frac{A^2}{\sqrt{2a}} \int_{x=-\infty}^{x=\infty} e^{-u^2} du$$

- Since $u = \sqrt{2a}\, x$, and $\sqrt{2a} > 0$, we know $u|_{x=-\infty} = -\infty$ and $u|_{x=\infty} = \infty$

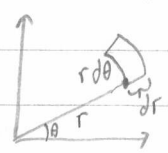
$$\Rightarrow \frac{A^2}{\sqrt{2a}} \int_{-\infty}^{\infty} e^{-u^2} du$$

- Let's separate out the $\int_{-\infty}^{\infty} e^{-u^2} du$. To evaluate this, we can use a trick I learned from my multivariable calculus class:
  - > It actually turns out to be easier to solve for the 3D version of the problem:

$$\int_{v=-\infty}^{v=\infty} \int_{u=-\infty}^{u=\infty} e^{-(u^2+v^2)} du\, dv = \iint_{uv\text{-plane}} e^{-(u^2+v^2)} dA$$

  - > Now, the above is still hard to do unless we realise we can use polar coordinates:

$r^2 = u^2 + v^2$, $\theta = \tan^{-1}\left(\frac{v}{u}\right)$, and $dA = r\, d\theta\, dr$



$dA = r\, d\theta\, dr$

  - > Thus:

$$\iint_{uv\text{-plane}} e^{-(u^2+v^2)} dA = \int_{r=0}^{r=\infty} \int_{\theta=0}^{\theta=2\pi} e^{-r^2} r\, d\theta\, dr = \int_0^{\infty} 2\pi \left(e^{-r^2} r\right) dr = 2\pi \int_0^{\infty} e^{-r^2} r\, dr$$

  - > Let $b = -r^2$, so $db = -2r\, dr \Rightarrow dr = -\frac{1}{2r} db$

$$\Rightarrow \iint_{uv\text{-plane}} e^{-(u^2+v^2)} dA = 2\pi \int_{r=0}^{r=\infty} -\frac{1}{2} e^{b}\, db = -\pi \int_{r=0}^{r=\infty} e^{b}\, db = -\pi \left[e^{b} \right]_{r=0}^{r=\infty}$$

$$\Rightarrow \iint_{uv\text{-plane}} e^{-(u^2+v^2)} dA = -\pi \left[e^{-r^2} \right]_{r=0}^{r=\infty} = -\pi \left[e^{-\infty} - e^{0} \right] = -\pi(-1)$$

$$\Rightarrow \iint_{uv\text{-plane}} e^{-(u^2+v^2)} dA = \pi$$

> Alright, now let $C = \int_{-\infty}^{\infty} e^{-u^2} du$, and plug that into $\iint_{uv\text{-plane}} e^{-(u^2+v^2)} dA = \pi$:

$$\iint_{uv\text{-plane}} e^{-(u^2+v^2)} dA = \int_{v=-\infty}^{v=\infty} \int_{u=-\infty}^{u=\infty} e^{-u^2} e^{-v^2} du\, dv = \pi$$

$$\Rightarrow \int_{v=-\infty}^{v=\infty} e^{-v^2} \left(\int_{u=-\infty}^{u=\infty} e^{-u^2} du \right) dv = \int_{v=-\infty}^{v=\infty} C e^{-v^2} dv = C \int_{v=-\infty}^{v=\infty} e^{-v^2} dv = \pi$$

> Now, just note that $\int_{v=-\infty}^{v=\infty} e^{-v^2} dv$ must be equal to $C = \int_{u=-\infty}^{u=\infty} e^{-u^2} du$, since all we did was put the variable v in place of u. Thus:

$$C \left(\int_{v=-\infty}^{v=\infty} e^{-v^2} dv \right) = C(C) = C^2 = \pi \Rightarrow C = \sqrt{\pi}$$

$$\Rightarrow \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$$

- Now that we know $\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$, we can plug that in:

$$\int_{-\infty}^{\infty} |\Psi(x,0)|^2 dx = \frac{A^2}{\sqrt{2a}} \int_{-\infty}^{\infty} e^{-u^2} du = \frac{A^2 \sqrt{\pi}}{\sqrt{2a}} = 1$$

$$\Rightarrow A^2 = \sqrt{\frac{2a}{\pi}} \Rightarrow \boxed{A = \left(\frac{2a}{\pi} \right)^{1/4}}$$

<u>Significance</u>

- This problem shows that the mathematics surrounding bell curves - typically associated with probability - is also very useful in quantum mechanics; further cementing the probabalistic nature of quantum mechanics.

1b Strategy

- We know that we can write $\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x,0)\, e^{ikx}\, dx$ so that $\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{i(kx-\omega t)} d$
- Also, it may be useful to recall that, for a free particle, $\omega = \frac{\hbar k^2}{2m}$
- If we combine the above two facts, getting to a solution should be straightforward algebra and calculus.

Solution

- First evaluate $\phi(k)$.

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x,0) e^{ikx} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A e^{-ax^2} e^{ikx} dx = \frac{1}{\sqrt{2\pi}} \cdot \left(\frac{2a}{\pi}\right)^{1/4} \int_{-\infty}^{\infty} e^{-ax^2} e^{ikx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \cdot \left(\frac{2a}{\pi}\right)^{1/4} \int_{-\infty}^{\infty} \cos(kx)\, e^{-ax^2} - i\sin(kx) e^{-ax^2} dx = \frac{1}{\sqrt{2\pi}} \left(\frac{2a}{\pi}\right)^{1/4} \left(\int_{-\infty}^{\infty} \cos(kx) e^{-ax^2} dx - \int_{-\infty}^{\infty} i\sin(kx) e^{-ax^2} d \right)$$

- We can note that $i\sin(kx)$ is an odd function while e^{-ax^2} is an even function, meaning that $\int_{-\infty}^{\infty} i\sin(kx) e^{-ax^2} dx = 0$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \left(\frac{2a}{\pi}\right)^{1/4} \int_{-\infty}^{\infty} \cos(kx) e^{-ax^2} dx$$

- Now, we can use the fact that $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ to rewrite:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \left(\frac{2a}{\pi}\right)^{1/4} \int_{-\infty}^{\infty} \frac{1}{2} e^{-ax^2} (e^{ikx} + e^{-ikx}) dx = \frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\right)\left(\frac{2a}{\pi}\right)^{1/4} \left(\int_{-\infty}^{\infty} e^{-(ax^2 - ikx)} dx + \int_{-\infty}^{\infty} e^{-(ax^2 + ikx)} dx \right)$$

- Let $y = \sqrt{a}\left[x - \frac{ik}{2a}\right]$ and $z = \sqrt{a}\left[x + \frac{ik}{2a}\right]$. Then, we can note that:
  - $y^2 - \left(\frac{(ik)^2}{4a}\right) = a\left(x - \frac{ik}{2a}\right)^2 - \left(\frac{-k^2}{4a}\right) = a\left(x^2 - \frac{ik}{a}x - \frac{k^2}{4a^2}\right) + \frac{k^2}{4a} = ax^2 - ikx - \frac{k^2}{4a} + \frac{k^2}{4a}$
    $\Rightarrow y^2 + \frac{k^2}{4a} = ax^2 - ikx$
  - $z^2 - \left(\frac{(ik)^2}{4a}\right) = a\left(x + \frac{ik}{2a}\right)^2 + \left(\frac{k^2}{4a}\right) = a\left(x^2 + \frac{ik}{a}x - \frac{k^2}{4a^2}\right) + \frac{k^2}{4a} = ax^2 + ikx - \frac{k^2}{4a} + \frac{k^2}{4a}$
    $\Rightarrow z^2 + \frac{k^2}{4a} = ax^2 + ikx$
- Now, let us rewrite:

$$\phi(k) = \frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\right)\left(\frac{2a}{\pi}\right)^{1/4} \left(\int_{-\infty}^{\infty} e^{-(y^2 + \frac{k^2}{4a})} dx + \int_{-\infty}^{\infty} e^{-(z^2 + \frac{k^2}{4a})} dx \right) = \frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\right)\left(\frac{2a}{\pi}\right)^{1/4} \left(\int_{-\infty}^{\infty} e^{-y^2} e^{-\frac{k^2}{4a}} dx + \int_{-\infty}^{\infty} e^{-z^2} e^{-\frac{k^2}{4a}} dx \right)$$

- We can note that $dy = \sqrt{a}\, dx$, so $dx = \frac{1}{\sqrt{a}} dy$. Additionally, $dz = \sqrt{a}\, dx$ so $dx = \frac{1}{\sqrt{a}} dz$

$$\phi(k) = \frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\right)\left(\frac{2a}{\pi}\right)^{1/4} \left(\int_{x=-\infty}^{x=\infty} e^{-y^2} \left(\frac{e^{-\frac{k^2}{4a}}}{\sqrt{a}}\right) dy + \int_{x=-\infty}^{x=\infty} e^{-z^2} \left(\frac{e^{-k^2/4a}}{\sqrt{a}}\right) dz \right)$$

$$= \frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\right)\left(\frac{2a}{\pi}\right)^{1/4} \left(\frac{e^{-\frac{k^2}{4a}}}{\sqrt{a}}\right) \left(\int_{x=-\infty}^{x=\infty} e^{-y^2} dy + \int_{x=-\infty}^{x=\infty} e^{-z^2} dz \right)$$

- Note that at $x=-\infty$, $y=\sqrt{2}\left(x-\frac{ik}{2a}\right)=-\infty$ and $z=\sqrt{2}\left(x+\frac{ik}{2a}\right)=-\infty$. Meanwhile, at $x=\infty$, $y=\sqrt{2}\left(x-\frac{ik}{2a}\right)=\infty$ and $z=\sqrt{2}\left(x+\frac{ik}{2a}\right)=\infty$. We can then rewrite the bounds of the integrals as:

$$\phi(k)=\frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\right)\left(\frac{2a}{\pi}\right)^{1/4}\left(\frac{e^{-\frac{k^2}{4a}}}{\sqrt{a}}\right)\left(\int_{-\infty}^{\infty}e^{-y^2}dy+\int_{-\infty}^{\infty}e^{-z^2}dz\right)$$

- As proved in part a, $\int_{-\infty}^{\infty}e^{-y^2}dy=\sqrt{\pi}$ and $\int_{-\infty}^{\infty}e^{-z^2}dz=\sqrt{\pi}$

$$\Rightarrow \phi(k)=\frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\right)\left(\frac{2a}{\pi}\right)^{1/4}\left(\frac{e^{-\frac{k^2}{4a}}}{\sqrt{a}}\right)\cdot 2\sqrt{\pi}$$

$$\Rightarrow \phi(k)=\frac{1}{\sqrt{2a}}\left(\frac{2a}{\pi}\right)^{1/4}e^{-\frac{k^2}{4a}} \Rightarrow \phi(k)=\frac{e^{-\frac{k^2}{4a}}}{(2\pi a)^{1/4}}$$

- Now, we can plug in and solve for $\Psi(x,t)$:

$$\Psi(x,t)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\phi(k)e^{i(kx-\omega t)}dk=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\left(\frac{e^{-\frac{k^2}{4a}}}{(2\pi a)^{1/4}}\right)e^{i\left(kx-\frac{\hbar k^2}{2m}t\right)}dk$$

$$=\frac{1}{\sqrt{2\pi}(2\pi a)^{1/4}}\int_{-\infty}^{\infty}e^{-\left(\frac{1}{4a}k^2-ikx+\frac{\hbar t}{2m}k^2\right)}dk=\frac{1}{\sqrt{2\pi}(2\pi a)^{1/4}}\int_{-\infty}^{\infty}e^{-\left[\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)k^2-(ix)k\right]}dk$$

- Just like before, complete the square by letting $u=\sqrt{i\frac{\hbar t}{2m}+\frac{1}{4a}}\left[k+\frac{-ix}{2\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}\right]$ so that

$$u^2-\left(\frac{(ix)^2}{4\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}\right)=\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)\left[k^2-\frac{ix}{i\frac{\hbar t}{2m}+\frac{1}{4a}}k-\frac{x^2}{4\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}\right]+\frac{x^2}{4\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}$$

$$\Rightarrow u^2+\frac{x^2}{4\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}=\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)k^2-(ix)k$$

- Thus:

$$\Psi(x,t)=\frac{1}{\sqrt{2\pi}(2\pi a)^{1/4}}\int_{-\infty}^{\infty}e^{-\left(u^2+\frac{x^2}{4\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}\right)}dk=\frac{1}{\sqrt{2\pi}(2\pi a)^{1/4}}\int_{-\infty}^{\infty}e^{-u^2}e^{-\frac{x^2}{4\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}}dk$$

- Note that $du=\sqrt{i\frac{\hbar t}{2m}+\frac{1}{4a}}\,dk$, and at $k=-\infty$, $u=-\infty$ and at $k=\infty$, $u=\infty$. Thus:

$$\Psi(x,t)=\frac{1}{\sqrt{2\pi}(2\pi a)^{1/4}}\int_{-\infty}^{\infty}e^{-u^2}\left(\frac{e^{-\frac{x^2}{4\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}}}{\sqrt{i\frac{\hbar t}{2m}+\frac{1}{4a}}}\right)du$$

$$\Rightarrow \Psi(x,t)=\left(\frac{1}{\sqrt{2\pi}(2\pi a)^{1/4}}\right)\left(\frac{e^{-\frac{x^2}{4\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}}}{\sqrt{i\frac{\hbar t}{2m}+\frac{1}{4a}}}\right)\int_{-\infty}^{\infty}e^{-u^2}du=\frac{1}{\sqrt{2\pi}(2\pi a)^{1/4}}\cdot\frac{e^{-\frac{x^2}{4\left(i\frac{\hbar t}{2m}+\frac{1}{4a}\right)}}}{\sqrt{i\frac{\hbar t}{2m}+\frac{1}{4a}}}\cdot\sqrt{\pi}$$

$$\Rightarrow \Psi(x,t)=\frac{1}{\sqrt{2}(2\pi a)^{1/4}}\left(\frac{1}{\sqrt{\frac{i\hbar t}{2m}+\frac{1}{4a}}}\cdot\frac{\sqrt{4a}}{\sqrt{4a}}\right)e^{-\left(\frac{x^2}{\frac{2i\hbar t}{m}+\frac{1}{a}}\cdot\frac{a}{a}\right)}$$

1c Strategy

- To find $|\Psi(x,t)|^2$, just multiply the wave function by its complex conjugate.
- To find the behaviour at $t=0$, substitute 0 in for t in the equation for $|\Psi(x,t)|^2$
- To find the behaviour after a long period of time has passed, find $\lim_{t\to\infty} |\Psi(x,t)|^2$

Solution

$$|\Psi(x,t)|^2 = \Psi^*(x,t)\,\Psi(x,t) = \left(\frac{2a}{\pi}\right)^{1/4}\left(\frac{1}{\sqrt{1-\frac{2i\hbar a t}{m}}}\right)e^{-\frac{ax^2}{1-\frac{2i\hbar a t}{m}}}\left(\frac{2a}{\pi}\right)^{1/4}\left(\frac{1}{\sqrt{1+\frac{2i\hbar a t}{m}}}\right)e^{-\frac{ax^2}{1+\frac{2i\hbar a t}{m}}}$$

$$|\Psi(x,t)|^2 = \sqrt{\frac{2a}{\pi}}\left(\frac{1}{\sqrt{\left(1-\frac{2i\hbar a t}{m}\right)\left(1+\frac{2i\hbar a t}{m}\right)}}\right)e^{-ax^2\left(\frac{1}{1-\frac{2i\hbar a t}{m}}+\frac{1}{1+\frac{2i\hbar a t}{m}}\right)}$$

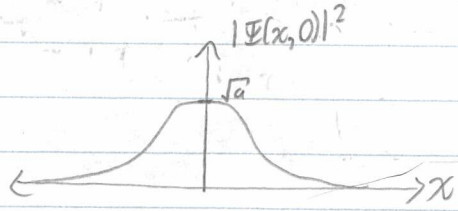
$$= \sqrt{\frac{2}{\pi}}\left(\sqrt{\frac{a}{1+\left(\frac{2\hbar a t}{m}\right)^2}}\right)e^{-ax^2\left(\frac{2}{1+\left(\frac{2\hbar a t}{m}\right)^2}\right)}$$

$$|\Psi(x,t)|^2 = \sqrt{\frac{2}{\pi}}\left(\sqrt{\frac{a}{1+\left(\frac{2\hbar a t}{m}\right)^2}}\right)e^{-2\left(\frac{a}{1+\left(\frac{2\hbar a t}{m}\right)^2}\right)x^2}$$

- Now, use the definition $w=\sqrt{\frac{a}{1+\left(\frac{2\hbar a t}{m}\right)^2}}$ to rewrite:

$$\boxed{|\Psi(x,t)|^2 = w\sqrt{\frac{2}{\pi}}\,e^{-2w^2x^2}}$$

- Note that, at $t=0$, $w=\sqrt{a}$, so $|\Psi(x,0)|^2 = \sqrt{\frac{2a}{\pi}}\,e^{-2ax^2}$. Graphing this, it roughly looks like:



- As t becomes larger, w becomes smaller, decreasing the maximum "peak value" of $|\Psi(x,t)|^2$. At the same time, for $e^{-w^2x^2}$, a decreased value of w would make it take a greater increase in x to decay the same amount in $|\Psi(x,t)|^2$. The combination of these two effects means that, as t increases, the graph of $|\Psi(x,t)|^2$ vs. x will visually "flatten," becoming very flat/delocalised for large t.

Significance

- The "flattening" of the graph of $|\Psi(x,t)|^2$ as t increases gives us a way of thinking about why various gaussian packets may experience natural dispersion as time increases.

$$\Rightarrow \Psi(x,t) = \frac{1}{\sqrt{2}\,(2\pi a)^{1/4}} \left(\frac{2\sqrt{a}}{\sqrt{\frac{2i\hbar a t}{m} + 1}} \right) e^{-\left(\frac{ax^2}{1 + \frac{2i\hbar a t}{m}}\right)}$$

- Let us define $\gamma \equiv \sqrt{1 + \frac{2i\hbar a t}{m}}$:

$$\Psi(x,t) = \frac{2(a^{1/2})}{(2^{3/4})(\pi^{1/4})(a^{1/4})} \left(\frac{1}{\gamma}\right) e^{-\frac{ax^2}{\gamma^2}} = \frac{(2^{1/4})(a^{1/4})}{(\pi^{1/4})} \left(\frac{1}{\gamma}\right) e^{-\frac{ax^2}{\gamma}}$$

$$\Rightarrow \boxed{\Psi(x,t) = \left(\frac{2a}{\pi}\right)^{1/4} \frac{1}{\gamma}\, e^{-\frac{ax^2}{\gamma^2}}}$$

<u>Significance</u>

- This problem is another example of how the equation for $\phi(k)$ in terms of $\Psi(x,0)$ and the equation for $\Psi(x,t)$ in terms of $\phi(k)$ allow us to find how the wavefunction evolves as time passes from just the shape of the wavefunction at time $t=0$ (assuming we know that we are describing a free particle)

1d Strategy

- We can primarily rely on the equation $\langle A\rangle = \int_{-\infty}^{\infty} \Psi^*(x,t)\,\hat{A}\,\Psi(x,t)\,dx$ where $\langle A\rangle$ is the expectation value of A and $\hat{A}$ is the proper operator to probe the system for information concerning A. We should anticipate $\langle x\rangle = 0$ due to symmetry.
  - > Then, just remember $\hat{p} = \frac{\hbar}{i}\frac{\delta}{\delta x}$ and $\hat{p}^2 = \left(\frac{\hbar}{i}\right)^2 \frac{\delta}{\delta x} = -\hbar^2 \frac{\delta}{\delta x}$
- $\langle x\rangle$ and $\langle x^2\rangle$ can be found simply with $\langle x\rangle = \int_{-\infty}^{\infty} x\,|\Psi(x,t)|^2 dx$ and $\langle x^2\rangle = \int_{-\infty}^{\infty} x^2 |\Psi(x,t)|^2 dx$
- Then, we can just use $\sigma_x^2 = \langle x^2\rangle - \langle x\rangle^2$ and $\sigma_p^2 = \langle p^2\rangle - \langle p\rangle^2$

Solution

$$\langle x\rangle = \int_{-\infty}^{\infty} x\,|\Psi(x,t)|^2 dx = \int_{-\infty}^{\infty} x\, w\sqrt{\frac{2}{\pi}}\, e^{-2w^2x^2} dx = w\sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} x e^{-2w^2x^2} dx = w\sqrt{\frac{2}{\pi}}\left[-\frac{1}{4w^2} e^{-2w^2x^2}\right]\Bigg|_{-\infty}^{\infty}$$

$$= w\sqrt{\frac{2}{\pi}}\left(-\frac{1}{4w^2}\right)\left[e^{-2w^2x^2}\right]\Bigg|_{-\infty}^{\infty} = -\frac{2}{4\pi w}\left[0-0\right] \Rightarrow \boxed{\langle x\rangle = 0}$$

- Now focus on finding $\langle x^2\rangle$

$$\langle x^2\rangle = \int_{-\infty}^{\infty} x^2 |\Psi(x,t)|^2 dx = \int_{-\infty}^{\infty} x^2 w\sqrt{\frac{2}{\pi}}\, e^{-2w^2x^2} dx = w\sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} x^2 e^{-2w^2x^2} dx$$

> Integration by parts tells us $\int_a^b u \frac{dv}{dx} dx = -\int_a^b v \frac{du}{dx} dx + [uv]\Big|_a^b$

$$= w\sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} x \frac{d}{dx}\left[-\frac{1}{4w^2} e^{-2w^2x^2}\right] dx = w\sqrt{\frac{2}{\pi}}\left(-\int_{-\infty}^{\infty} -\frac{1}{4w^2} e^{-2w^2x^2} \cdot \frac{dx}{dx} dx + \left[-\frac{x}{4w^2 e^{-2w^2x^2}}\right]\Bigg|_{-\infty}^{\infty}\right.$$

← denominator grows faster than numerator

$$= w\sqrt{\frac{2}{\pi}}\left(\frac{1}{4w^2}\right)\left(\int_{-\infty}^{\infty} e^{-2w^2x^2} dx + [0-0]\right) = \frac{\sqrt{2}}{4\sqrt{\pi}\, w} \int_{-\infty}^{\infty} e^{-2w^2x^2} dx$$

> Now, let $u = \sqrt{2}\,wx$ so $du = \sqrt{2}\,w\,dx \Rightarrow dx = \frac{du}{\sqrt{2}\,w}$

$$= \frac{1}{4w}\sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} e^{-u^2} \cdot \frac{1}{\sqrt{2}\,w} du = \frac{1}{4w^2}\sqrt{\frac{1}{\pi}} \int_{-\infty}^{\infty} e^{-u^2} du$$

> As discussed in part a, $\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$, so

$$\langle x^2\rangle = \frac{1}{4w^2}\sqrt{\frac{1}{\pi}}\sqrt{\pi} \Rightarrow \boxed{\langle x^2\rangle = \frac{1}{4w^2}}$$

- Now focus on finding $\langle p\rangle$

$$\langle p\rangle = \int_{-\infty}^{\infty} \Psi^*(x,t)\left(\frac{\hbar}{i}\right)\frac{\delta \Psi(x,t)}{\delta x} dx = \int_{-\infty}^{\infty} \left(\frac{2a}{\pi}\right)^{1/4}\left(\frac{1}{\sqrt{1-\frac{2i\hbar a t}{m}}}\right) e^{-\frac{ax^2}{1-\frac{2i\hbar a t}{m}}} \left(\frac{\hbar}{i}\right)\left(\frac{2a}{\pi}\right)^{1/4}\left(\frac{1}{\sqrt{1+\frac{2i\hbar a t}{m}}}\right)\frac{\delta}{\delta x}\left[e^{-\frac{ax^2}{1-\frac{2i\hbar a t}{m}}}\right] dx$$

$$= \sqrt{\frac{2a}{\pi}}\left(\frac{1}{\sqrt{1+\left(\frac{2\hbar a t}{m}\right)^2}}\right)\left(\frac{\hbar}{i}\right)\int_{-\infty}^{\infty} e^{-\frac{ax^2}{1-\frac{2i\hbar a t}{m}}} e^{-\frac{ax^2}{1+\frac{2i\hbar a t}{m}}} \cdot -\frac{2ax}{1+\frac{2i\hbar a t}{m}} dx$$

$$= \sqrt{\frac{2}{\pi}}(w)\left(\frac{\hbar}{i}\right)\int_{-\infty}^{\infty} e^{-ax^2\left(\frac{1+\frac{2i\hbar a t}{m}+1-\frac{2i\hbar a t}{m}}{1+\left(\frac{2\hbar a t}{m}\right)^2}\right)} \cdot -\frac{2ax}{1+\frac{2i\hbar a t}{m}} dx = \left(\frac{w\hbar}{i}\right)\sqrt{\frac{2}{\pi}}\left(-\frac{2a}{1+\frac{2i\hbar a t}{m}}\right)\int_{-\infty}^{\infty} e^{-2x^2\left(\frac{a}{1+\left(\frac{2\hbar a t}{m}\right)^2}\right)} \cdot x\,dx$$

$$= \frac{w\hbar}{i}\sqrt{\frac{2}{\pi}}\left(-\frac{2a}{1+\frac{2i\hbar at}{m}}\right)\int_{-\infty}^{\infty} e^{-2w^2x^2}\cdot x\,dx = \frac{w\hbar}{i}\sqrt{\frac{2}{\pi}}\left(-\frac{2a}{1+\frac{2i\hbar at}{m}}\right)\left[\frac{1}{4w^2}e^{-2w^2x^2}\right]_{-\infty}^{\infty}$$

$$= \frac{w\hbar}{i}\sqrt{\frac{2}{\pi}}\left(-\frac{2a}{1+\frac{2i\hbar at}{m}}\right)[0-0] \Rightarrow \boxed{\langle p\rangle = 0}$$

- We may finally focus on the most ardvous expectation value: $\langle p^2\rangle$

$$\langle p^2\rangle = \int_{-\infty}^{\infty}\Psi^*(x,t)\,(-\hbar^2)\frac{\delta^2\Psi(x,t)}{\delta x^2}\,dx = \int_{-\infty}^{\infty}\left(\frac{2a}{\pi}\right)^{\frac{1}{4}}\left(\frac{1}{\sqrt{1-\frac{2i\hbar at}{m}}}\right)e^{-\frac{ax^2}{1-\frac{2i\hbar at}{m}}}(-\hbar^2)\left(\frac{2a}{\pi}\right)^{\frac{1}{4}}\left(\frac{1}{\sqrt{1+\frac{2i\hbar at}{m}}}\right)\frac{\delta^2}{\delta x^2}\left[e^{-\frac{ax^2}{1+\frac{2i\hbar at}{m}}}\right]dx$$

$$= \sqrt{\frac{2}{\pi}}\left(\frac{\sqrt{a}}{\sqrt{1+\left(\frac{2\hbar at}{m}\right)^2}}\right)(-\hbar^2)\int_{-\infty}^{\infty} e^{-\frac{ax^2}{1-\frac{2i\hbar at}{m}}}\cdot\frac{\delta}{\delta x}\left[-\frac{2a}{1+\frac{2i\hbar at}{m}}\cdot x e^{-\frac{ax^2}{1+\frac{2i\hbar at}{m}}}\right]dx$$

$$= -\hbar^2 w\sqrt{\frac{2}{\pi}}\left(-\frac{2a}{1+\frac{2i\hbar at}{m}}\right)\int_{-\infty}^{\infty} e^{-\frac{ax^2}{1-\frac{2i\hbar at}{m}}}\cdot\left[x\cdot -\frac{2ax}{1+\frac{2i\hbar at}{m}}e^{-\frac{ax^2}{1+\frac{2i\hbar at}{m}}} + e^{-\frac{ax^2}{1+\frac{2i\hbar at}{m}}}\right]dx$$

$$= w\hbar^2\sqrt{\frac{2}{\pi}}\left(\frac{2a}{1+\frac{2i\hbar at}{m}}\right)\int_{-\infty}^{\infty} e^{-\frac{ax^2}{1-\frac{2i\hbar at}{m}}}e^{-\frac{ax^2}{1+\frac{2i\hbar at}{m}}}\left[-\frac{2ax^2}{1+\frac{2i\hbar at}{m}}+1\right]dx$$

$$= w\hbar^2\sqrt{\frac{2}{\pi}}\left(\frac{2a}{1+\frac{2i\hbar at}{m}}\right)\int_{-\infty}^{\infty} e^{-x^2 a\left(\frac{2}{1+\left(\frac{2\hbar at}{m}\right)^2}\right)}\left[1-\frac{2ax^2}{1+\frac{2i\hbar at}{m}}\right]dx$$

$$= w\hbar^2\sqrt{\frac{2}{\pi}}\left(\frac{2a}{1+\frac{2i\hbar at}{m}}\right)\int_{-\infty}^{\infty} e^{-2w^2x^2}\left(1-\frac{2ax^2}{1+\frac{2i\hbar at}{m}}\right)dx = w\hbar^2\left(\frac{2a}{1+\frac{2i\hbar at}{m}}\right)\left[\int_{-\infty}^{\infty} e^{-2w^2x^2}dx - \int_{-\infty}^{\infty} e^{-2w^2x^2}\cdot\frac{2ax^2}{1+\frac{2i\hbar at}{m}}dx\right.$$

> First let's focus on solving for $\int_{-\infty}^{\infty} e^{-2w^2x^2}dx$

- We'll have $u = \sqrt{2}\,wx$ so $du = \sqrt{2}\,w\,dx \Rightarrow dx = \frac{1}{\sqrt{2}w}du$. Furthermore at $x=\infty$, $u=\infty$ and $x=-\infty$ has $u=-\infty$.

$$\int_{-\infty}^{\infty} e^{-2w^2x^2}dx = \int_{-\infty}^{\infty} e^{-u^2}\cdot\frac{1}{\sqrt{2}w}du = \frac{1}{\sqrt{2}w}\int_{-\infty}^{\infty} e^{-u^2}du$$

- As discussed in part a, $\int_{-\infty}^{\infty} e^{-u^2}du = \sqrt{\pi}$

$$\int_{-\infty}^{\infty} e^{-2w^2x^2}dx = \sqrt{\frac{\pi}{2}}\cdot\frac{1}{w}$$

> Next, we're prepared to focus on the $\int_{-\infty}^{\infty}\frac{2ax^2}{1+\frac{2i\hbar at}{m}}e^{-2w^2x^2}dx$

- Integration by parts tells us $\int_a^b u\frac{dv}{dx}dx = -\int_a^b v\frac{du}{dx}dx + [uv]\Big|_a^b$

$$\int_{-\infty}^{\infty}\frac{2ax^2}{1+\frac{2i\hbar at}{m}}e^{-2w^2x^2} = \int_{-\infty}^{\infty}\frac{2ax}{1+\frac{2i\hbar at}{m}}\cdot xe^{-2w^2x^2}dx = \int_{-\infty}^{\infty}\left(\frac{2ax}{1+\frac{2i\hbar at}{m}}\right)\cdot\frac{d}{dx}\left[-\frac{1}{4w^2}e^{-2w^2x^2}\right]dx$$

$$= -\int_{-\infty}^{\infty}\frac{1}{4w^2}e^{-2w^2x^2}\cdot\frac{d}{dx}\left[\frac{2ax}{1+\frac{2i\hbar at}{m}}\right]dx + \left[-\frac{2ax}{1+\frac{2i\hbar at}{m}}\left(\frac{1}{4w^2}\right)e^{-2w^2x^2}\right]_{-\infty}^{\infty}$$

$$= \frac{2a}{1+\frac{2i\hbar at}{m}}\left(\frac{1}{4w^2}\right)\int_{-\infty}^{\infty} e^{-2w^2x^2}\,dx + \left[\lim_{x\to\infty} - \frac{2ax}{(4w^2)(1+\frac{2i\hbar at}{m})e^{2w^2x^2}} + \lim_{x\to-\infty} \frac{2ax}{(4w^2)(1+\frac{2i\hbar at}{m})e^{2w^2x^2}}\right]$$

- As already proven, $\int_{-\infty}^{\infty} e^{-2w^2x^2}\,dx = \sqrt{\frac{\pi}{2}}\cdot\frac{1}{w}$

$$= \frac{2a}{(2w^2)(1+\frac{2i\hbar at}{m})}\left(\sqrt{\frac{\pi}{2}}\cdot\frac{1}{w}\right) + [0-0]$$

$$\Rightarrow \int_{-\infty}^{\infty} e^{-2w^2x^2}\cdot\frac{2ax^2}{1+\frac{2i\hbar at}{m}}\,dx = \frac{a}{(2w^2)(1+\frac{2i\hbar at}{m})}\left(\sqrt{\frac{\pi}{2}}\cdot\frac{1}{w}\right)$$

> We'll plug both of our results back into the expression for $\langle p^2\rangle$:

$$\langle p^2\rangle = w\hbar^2\sqrt{\frac{2}{\pi}}\left(\frac{2a}{1+\frac{2i\hbar at}{m}}\right)\left[\left(\sqrt{\frac{\pi}{2}}\cdot\frac{1}{w}\right) - \left(\sqrt{\frac{\pi}{2}}\cdot\frac{1}{w}\right)\left(\frac{a}{(2w^2)(1+\frac{2i\hbar at}{m})}\right)\right]$$

$$= w\hbar^2\sqrt{\frac{2}{\pi}}\left(\sqrt{\frac{\pi}{2}}\cdot\frac{1}{w}\right)\left(\frac{2a}{1+\frac{2i\hbar at}{m}}\right)\left[1 - \frac{a}{\frac{2a}{(1+\frac{2i\hbar at}{m})(1-\frac{2i\hbar at}{m})}(1+\frac{2i\hbar at}{m})}\right]$$

$$= a\hbar^2\left(\frac{2}{1+\frac{2i\hbar at}{m}}\right)\left(1 - \frac{a(1-\frac{2i\hbar at}{m})}{2a}\right) = a\hbar^2\left(\frac{2}{1+\frac{2i\hbar at}{m}}\right)\left(1 - \frac{1-\frac{2i\hbar at}{m}}{2}\right)$$

$$= a\hbar^2\left(\frac{2}{1+\frac{2i\hbar at}{m}}\right)\left(\frac{2-1+\frac{2i\hbar at}{m}}{2}\right) = a\hbar^2\left(\frac{2}{1+\frac{2i\hbar at}{m}}\right)\left(\frac{1+\frac{2i\hbar at}{m}}{2}\right)$$

$$\Rightarrow \boxed{\langle p^2\rangle = a\hbar^2}$$

- Now, it becomes incredibly straightforward to find σ_x and σ_p

$$\sigma_x^2 = \langle x^2\rangle - \langle x\rangle^2 = \frac{1}{4w^2} - 0^2 \Rightarrow \boxed{\sigma_x = \frac{1}{2w}}$$

$$\sigma_p^2 = \langle p^2\rangle - \langle p\rangle^2 = a\hbar^2 \Rightarrow \boxed{\sigma_p = \hbar\sqrt{a}}$$

Significance

- These calculations show that for a simple Gaussian wave packet with $\Psi(x,0) = Ae^{-ax^2}$, the expectation value of the momentum is always zero, meaning the wave packet must have a more complicated form if it is to have some initial momentum.
- Additionally, we can see that even if $\langle x\rangle = 0$ and $\langle p\rangle = 0$, it is not necessarily the case that $\langle x^2\rangle$ or $\langle p^2\rangle$ are zero.